

Gopi Krishna Erabati

Machine Learning Engineer – Perception – PhD

Berlin, Germany

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Summary

Perception and Machine Learning Engineer with a PhD in Automation and Robotics and 7+ years building Computer Vision, Deep Learning, and 3D Perception systems for Autonomous Driving. Specialist in LiDAR-camera-radar sensor fusion, semantic segmentation, 3D object detection, occupancy prediction, BEV prediction, Vision-Language-Action (VLA) models, and real-time deployment on NVIDIA edge GPUs. Experienced in Python, PyTorch, OpenCV, TensorRT, and multimodal and multitask deep learning for robotics and autonomous vehicles. Author of 13+ publications at ICRA, WACV, T-IV, Neurocomputing and CVIU.

Work Experience

Sensmore GmbH

Berlin, Germany

Software Engineer, Perception

Jan. 2025 – Present

- Lead ML and Data initiatives for the perception stack of off-highway autonomous vehicles in harsh environments
- Developed a multimodal Vision-Language-Action (VLA) model for physical AI, integrating camera, LiDAR, and language inputs for autonomous driving
- Developed a real-time multimodal, multitask semantic occupancy prediction network using LiDAR, radar, and camera fusion for off-highway autonomous vehicle perception
- Improved model performance and throughput by 15% and reduced data-loading latency by 20% through pipeline optimization, accelerating training and inference workflows

Institute of Systems and Robotics

Coimbra, Portugal

Applied ML Research Engineer

Jan. 2019 – Dec. 2024

- Designed and implemented deep learning models for autonomous driving scene understanding, including 3D object detection, LiDAR semantic segmentation, panoptic perception, and sensor fusion under real world constraints such as sparse data, adverse weather, small objects and real-time deployment
- Improved performance (3× speedup) and robustness in adverse weather for small-object detection by developing novel LiDAR-based and LiDAR-camera fusion 3D object detection approaches
- Achieved real-time inference (112 FPS) on NVIDIA Jetson edge GPU for panoptic driving perception and developed LiDAR-based 3D semantic segmentation, improving accuracy for distant objects in sparse LiDAR point clouds using retention

LAAS – CNRS

Toulouse, France

Computer Vision Research Intern

Feb. 2018 – July 2018

- Designed a novel approach for 3D object detection and relative localization using a 3D sensor embedded on a mobile robot and Integrated the proposed module with ROS
- Designed a novel approach for forest fire mapping from low altitude aerial imagery

Technical Skills

Perception & ML: Object Detection, Semantic Segmentation, Occupancy Prediction, BEV Perception, LiDAR-Camera-Radar Fusion, Vision-Language-Action (VLA) Models, Transformers, CNN

Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, PCL, NumPy, ONNX, TensorRT, ROS, mmdetection

Programming & Tools: Python, C++, MATLAB, Git, Docker, CI/CD

Languages: English (Fluent), Telugu (Native), Hindi (Fluent), Portuguese (Elementary)

Education

University of Coimbra

Coimbra, Portugal

Doctor of Philosophy (PhD)

Sept. 2019 – Feb. 2025

- Thesis: *Learning to Perceive: Scene Understanding for Autonomous Driving*
- Part of EU H2020 Marie Skłodowska-Curie Actions Innovative Training Network ACHIEVE
- Supervisor: Prof. Helder Araujo

Université de Dijon

Le Creusot, France

M.Sc. in Computer Vision (VIBOT)

Sept. 2016 – Aug. 2018

- Thesis: *3D object detection and relative localization using a 3D sensor embedded on a mobile robot*
- Courses: Visual Perception, Machine Learning, Probabilistic Robotics, Autonomous Robotics
- Grade: 15.0/20.0 (Rank: 2/16)

- Thesis: *Development of quadcopter for search and rescue in natural disasters*
- Grade: 89.8 % (Rank: 1/66)
- Received Gold Medal for excellence in academics

Projects and Training

Projects (More details: <https://gopi-erabati.github.io/projects/>)

- Human Activity Recognition in Videos
- Mapping, Autonomous Navigation and Localization of Turtlebot using ROS
- Development of Computer Vision Toolbox in C++ and MATLAB using OpenCV
- Development of 3D Scanner using Kinect and PCL
- Development of Face Recognition software using PCA

Training

- Attended **Oxford Machine Learning Summer School** (OxML 2023) at University of Oxford, UK
- Deep Learning Specialization taught by Prof. Andrew Ng on Coursera
- Attended AI-DLDA 2020 International Summer School on Artificial Intelligence at Università di Udine, Italy
- Presented at ICRA 2025, WACV 2023, ICPRW 2022, ROBOVIS 2024, RECPAD 2021, DICTA 2020, ICDSC 2019

Selected Publications

SCAM-P: Spatial Channel Attention Module for Panoptic Driving Perception

G. K. Erabati, H. Araujo

2025 IEEE International Conference on Robotics and Automation (ICRA), 2025

RetFormer: Embracing Point Cloud Transformer with Retentive Network

G. K. Erabati, H. Araujo

IEEE Transactions on Intelligent Vehicles (IEEE T-IV), 2024

SRFDet3D: Sparse Region Fusion based 3D Object Detection

G. K. Erabati, H. Araujo

Neurocomputing 593, 2024

RetSeg3D: Retention-based 3D Semantic Segmentation for Autonomous Driving

G. K. Erabati, H. Araujo

Computer Vision and Image Understanding (CVIU), 2024

Li3DeTr: A LiDAR Based 3D Detection Transformer

G. K. Erabati, H. Araujo

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2023

MSF3DDETR: Multi-Sensor Fusion 3D Detection Transformer for Autonomous Driving

G. K. Erabati, H. Araujo

ICPR 2022 workshop on Deep Learning for Visual Detection and Recognition (DLVDR), 2022

DDet3D: Embracing 3D Object Detector with Diffusion

G. K. Erabati, H. Araujo

Applied Intelligence, 2024

Honors & Awards

FCT PhD Scholarship Grant, Fundação para a Ciência e a Tecnologia, Portugal, 2021

Marie Skłodowska–Curie Fellowship Grant, European Commission, 2019

Junior Research Fellowship Grant, Defence Research and Development Organization (DRDO), India, 2014

Gold Medalist, Kakatiya University, India, 2013

Featured in the Roll of Honor Board, Kakatiya Institute of Technology and Science, India, 2013